

Distance Metrics for Classification

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closeness/similarity b/w two input points

↳ Influences the decision boundaries of algorithms

① Euclidean Distance

$\vec{x}_1: x_{11}, x_{21}, \dots, x_{p1}$

$\vec{x}_2: x_{12}, x_{22}, \dots, x_{p2}$

$\vec{x}$
← p-dim →

(categorical)
 y

$\hat{y}$
predicted

$\vec{x}_1$ x_{11} x_{21} $\dots$ x_{p1}

y_1

$\hat{y}_1$

$\vec{x}_2$ x_{12} x_{22} $\dots$ x_{p2}

y_2

$\hat{y}_2$

$\vdots$

$\vdots$

$\vdots$

$\vdots$

$\vdots$

$\vdots$

$\vec{x}_N$ x_{1N} x_{2N} $\dots$ x_{pN}

y_N

$\hat{y}_N$

↓

↓

↓

features/attributes

$$\text{Euclidean distance} = \sqrt{\sum_{i=1}^P (x_{i1} - x_{i2})^2}$$

eg. $x_1 = (1, 2), \quad x_2 = (4, 6)$

$$\text{Euclidean distance} = \sqrt{(1-4)^2 + (2-6)^2} = \sqrt{9+16} = \boxed{5}$$

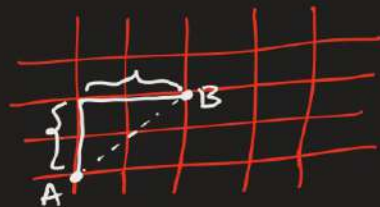
eg. $x_1 = (1, 1, 2, 3), \quad x_2 = (2, 4, 1, 5)$

$$\begin{aligned} \text{Euclidean distance} &= \sqrt{(1-2)^2 + (1-4)^2 + (2-1)^2 + (3-5)^2} \\ &= \sqrt{1+9+1+4} = \boxed{\sqrt{15}} \end{aligned}$$

② Manhattan Distance (Taxicab or L1 norm)

$$\vec{x}_1: x_{11}, x_{21}, \dots, x_{p1}$$

$$\vec{x}_2: x_{12}, x_{22}, \dots, x_{p2}$$



$$\text{Manhattan Distance} = \sum_{i=1}^p |x_{i1} - x_{i2}|$$

eg. $x_1 = (1, 2), \quad x_2 = (4, 6)$

$$\text{Manhattan Distance} = |1-4| + |2-6| = 3+4 = \boxed{7}$$

eg. $x_1 = (1, 1, 2, 3), \quad x_2 = (2, 4, 1, 5)$

$$\begin{aligned} \text{Manhattan Distance} &= |1-2| + |1-4| + |2-1| + |3-5| \\ &= 1+3+1+2 = \boxed{7} \end{aligned}$$

③ Minkowski Distance

$$\vec{x}_1: x_{11}, x_{21}, \dots, x_{p1}$$

$$\vec{x}_2: x_{12}, x_{22}, \dots, x_{p2}$$

$$\text{Minkowski Distance} = \left(\sum_{i=1}^p |x_{i1} - x_{i2}|^t \right)^{1/t}$$

$t=1$:- Manhattan Distance

$t=2$:- Euclidean Distance

eg: $x_1 = (1, 2), \quad x_2 = (4, 6)$

$$\text{Minkowski Distance } (t=3) = (|1-4|^3 + |2-6|^3)^{1/3} \approx \boxed{4.498}$$

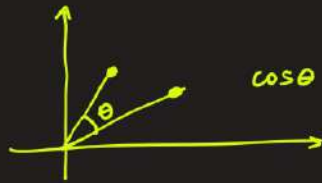
eg: $x_1 = (1, 1, 2, 3), \quad x_2 = (2, 4, 1, 5)$

$$\text{Minkowski Distance } (t=3) = (|1-2|^3 + |1-4|^3 + |2-1|^3 + |3-5|^3)^{1/3} \approx \boxed{3.332}$$

④ Cosine Similarity

use it when the magnitude of the vector doesn't matter

(where the input features are normalized to same scale)



$$\vec{x}_1: x_{11}, x_{21}, \dots, x_{p1}$$

$$\vec{x}_2: x_{12}, x_{22}, \dots, x_{p2}$$

$$\text{Cosine Similarity} = \frac{\sum_{i=1}^P x_{i1} * x_{i2}}{\sqrt{\sum_{i=1}^P x_{i1}^2} \sqrt{\sum_{i=1}^P x_{i2}^2}}$$

$$\vec{a} \cdot \vec{b} = |\vec{a}| \cdot |\vec{b}| \cdot \cos \theta$$

$$\cos \theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}| \cdot |\vec{b}|}$$

eg: $x_1 = (1, 2), \quad x_2 = (4, 6)$

$$\text{Cosine Similarity} = \frac{1 \times 4 + 2 \times 6}{\sqrt{1^2 + 2^2} \cdot \sqrt{4^2 + 6^2}} = \frac{16}{\sqrt{5} \cdot \sqrt{52}} \approx \boxed{0.992}$$

eg: $x_1 = (1, 1, 2, 3), \quad x_2 = (2, 4, 1, 5)$

$$\text{Cosine Similarity} = \frac{1 \times 2 + 1 \times 4 + 2 \times 1 + 3 \times 5}{\sqrt{1^2 + 1^2 + 2^2 + 3^2} \cdot \sqrt{2^2 + 4^2 + 1^2 + 5^2}} = \frac{23}{\sqrt{15} \cdot \sqrt{46}} \approx \boxed{0.876}$$

- If distance $\uparrow$ points are far from each other
- If similarity $\uparrow$ points are nearer to each other

⑤ Chebyshev Distance

↳ greatest of their distances along any coordinate dimension.

$$\vec{x}_1: x_{11}, x_{21}, \dots, x_{p1} \quad \vec{x}_2: x_{12}, x_{22}, \dots, x_{p2}$$

$$\text{Chebyshev Distance} = \max_i |x_{i1} - x_{i2}|$$

eg: $x_1 = (1, 2), \quad x_2 = (4, 6)$

$$\text{Chebyshev distance} = \max(|1-4|, |2-6|) = \max(3, 4) = \boxed{4}$$

eg: $x_1 = (1, 1, 2, 3), \quad x_2 = (2, 4, 1, 5)$

$$\begin{aligned} \text{Chebyshev Distance} &= \max(|1-2|, |1-4|, |2-1|, |3-5|) \\ &= \max(1, 3, 1, 2) = \boxed{3} \end{aligned}$$

⑥ Jaccard Index

↳ is used for comparing the similarity and diversity of sample sets

A and B are two sets. then

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|} = \frac{|A \cap B|}{|A| + |B| - |A \cap B|}$$

eg. $A = \{1, 2, 3, 4\}, \quad B = \{3, 4, 5, 6\}$

$$A \cap B = \{3, 4\}, \quad A \cup B = \{1, 2, 3, 4, 5, 6\}$$

$$\therefore J(A, B) = \frac{|A \cap B|}{|A \cup B|} = \frac{2}{6} \approx \boxed{0.33}$$

• Jaccard index is a similarity score

⑦ Hamming distance

↳ measures the number of positions at which the corresponding characters are different b/w two strings of equal length.

x, y are two strings, then:

$$d_{\text{Hamming}}(x, y) = \sum_{i=1}^n 1(x_i \neq y_i)$$

where $1(x_i \neq y_i)$ is an indicator function that is 1 if x and y i^{th} character do not match

eg. $x = \underline{1}\underline{1}\underline{0}\underline{0}\underline{1}\underline{0}\underline{1}$, $y = \underline{1}\underline{0}\underline{0}\underline{1}\underline{0}\underline{0}\underline{1}$

$$d_{\text{Hamming}} = \underset{\substack{\text{pos} \\ 2}}{1} + \underset{\substack{\text{pos} \\ 4}}{1} + \underset{\substack{\text{pos} \\ 5}}{1} = \boxed{3}$$

= x =

K-Nearest Neighbors

- ↳ Supervised classification algorithm
- ↳ It relies on distance metrics to determine the "closeness" of data points.
- ↳ It predicts the class of a given point based on the majority class of k of its nearest neighbors
- ↳ k is a positive integer, typically not very large.
- ↳ If $k=1$, then the new data point is just assigned to the class of its nearest neighbour.

How KNN works?

- (1) choose the Number of Neighbors (k)
- (2) Calculate distances - calculate the distance of the new data point with all the data points in training set.
- (3) Find the K -nearest neighbors - distance , similarity
 ↓ ↓
 smallest k largest k
 values values
- (4) Vote for labels.

Choosing the right value of k

Small k :- Makes the model sensitive to noise. The decision boundary is more intricate

Large k :- Makes the boundary smoother and the model more resistant to outliers, but it will blur the class distinctions.

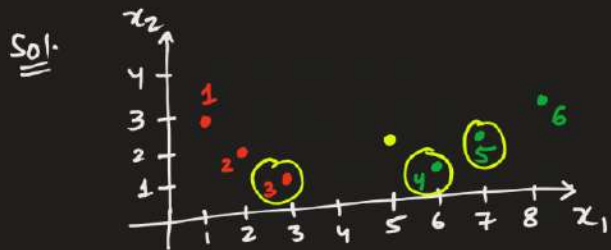
Choosing the Distance Metric

- Euclidean:- Good for low dimensional data
- Manhattan:- Good for high dimensional data, or for data with underlying grid-like structure
- Cosine:- useful when the magnitude of data vectors is not relevant.
- Jaccard index:- When a feature contains sets as data
- Hamming distance:- when a feature contains strings of equal length as data.

eg.

Instance	x_1	x_2	actual class y (0/1)
1	1	3	0
2	2	2	0
3	3	1	0
4	6	1	1
5	7	2	1
6	8	3	1

Calculate the predicted label for the data point $x = (5, 2)$ using KNN algorithm with $k=3$.



Instance	x_1	x_2	actual class y (0/1)	Euclidean distance from (5, 2)
1	1	3	0	$\sqrt{(1-5)^2 + (3-2)^2} = \sqrt{17}$
2	2	2	0	$\sqrt{(2-5)^2 + (2-2)^2} = \sqrt{9}$
3	3	1	0 ←	$\sqrt{(3-5)^2 + (1-2)^2} = \sqrt{5}$ (3)
4	6	1	1 ←	$\sqrt{(6-5)^2 + (1-2)^2} = \sqrt{2}$ (1)
5	7	2	1 ←	$\sqrt{(7-5)^2 + (2-2)^2} = \sqrt{4}$ (2)
6	8	3	1	$\sqrt{(8-5)^2 + (3-2)^2} = \sqrt{10}$

predicted label = majority (0, 1, 1) = 1